MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11746280
COEFFICIENT-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Coefficient-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746280	Circuit Card Assembly, Coefficient-1

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is one (1) or two (2) single ended printed circuit boards consisting of multipliers necessary to provide the variable DC voltage signals to the visual image generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746280.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +15 VDC
b. -15 VDC
c. +15 VDC Variable
d. -15 VDC Variable

3.2.1.2 Multipliers.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on Drawing 11746280 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746280 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.2.1 Test equipment.- (See Figure 1)

4.2.1.1 Power Supply (PS1 & PS2).- The power supplies shall be 15 ± 0.05 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS3 & PS4).- The power supplies shall be 15 ± 0.001 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Dummy Loads.- The dummy loads shall be 10K, 1/4 W, 1% resistors.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall met with the following power supply voltages.

a. +15.00 ± 0.05 VDC
b. -15.00 ± 0.05 VDC
c. 0.001 to +15.000 ± 0.001 VDC
d. 0.001 to -15.000 ± 0.001 VDC

TABLE I

PINS

INPUT		OUTPUT
33	32	41
26	31	30
15	23	21
1	2	7

VOLTAGES (VDC)

INPUT (± 0.001)		OUTPUT (± 0.1)
Col A	Col B	Col C
0.000	0.000	0
1.000	5.000	0.5
-5.000	-5.000	2.5
10.000	-5.000	-5.0
-10.000	10.000	-10.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

NOTE:

a. Each multiplier has two (2) variable inputs and one variable output, therefore, when the voltages listed in Col A and Col B are sequencially applied to the inputs, the results shall be the corresponding output as shown in Col C.

b. The output tolerance during extreme temperature testing shall be ± 0.3 VDC.

c. When the voltage on pin 7 is -10.0 VDC, the voltage on pin 9 shall be -4.35 VDC.

TABLE II

PINS

INPUT				OUTPUT
36	35	39	38	37
24	28	27	25	29
10	16	19	18	17
2	3	6	5	4

VOLTAGE (VDC)

INPUT (± 0.001)				OUTPUT (± 0.1)
Col A	Col B	Col C	Col D	Col E
0.000	0.000	0.000	0.000	0.0
1.000	5.000	5.000	1.000	-1.6
10.000	1.000	-1.000	-10.000	-8.1
10.000	5.000	10.000	5.000	2.5
-10.000	-1.000	-10.000	-1.000	8.1

NOTE:

a. Each multiplier has four variable inputs and one (1) variable output, therefore, when the voltages listed in Columns A, B, C and D are sequencially applied to the inputs, the results shall be the corresponding output as shown in Col E.

b. The output tolerances during extreme temperature testing shall be ± 0.3 VDC.

4.2.2.2 Test Set-Up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3 Multiplier Test.- When specific voltage values of between ± 10 VDC are applied to the inputs of the multiplier the corresponding output shall be as shown in Tables I and II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

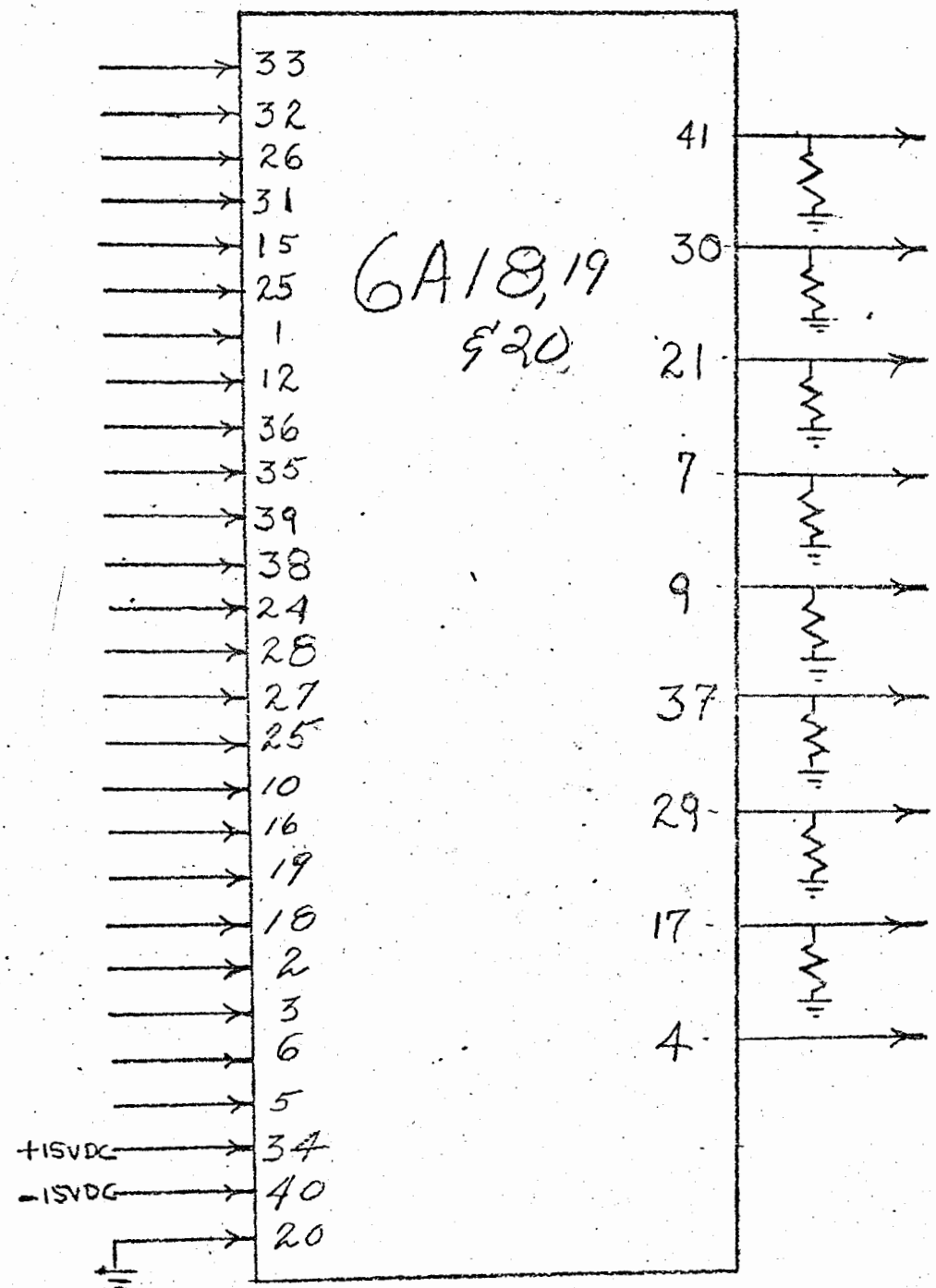


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746280.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A611

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.